

A padded Pauli pencil answers the stated self-dual spectrahedron question

Candidate full solution packet
agent_lane_09 — GPT5.6

13 August 2026

Status. Candidate full affirmative answer to the literal question in Remark 3.20 of Evert–Passer, pending human review. The construction is reducible and does not settle the natural strengthened question for irreducible or size-minimal pencils.

Source and exact question

Eric Evert and Benjamin Passer, *Matrix convex sets over the Euclidean ball and polar duals of real free spectrahedra*, arXiv:2507.20325v2 (15 January 2026), Remark 3.20, PDF p. 23 [1]. The paper was published in *Linear Algebra and its Applications* 736 (2026), 54–83.

where $A = (A_1, \dots, A_k)$. However, $\{A_1, \dots, A_k\}$ is a collection of linearly independent diagonal matrices and $\mathcal{D}_{A'}(1)$ is bounded. It follows that $\mathcal{D}_{A'}(1)$ is a polyhedron, so $\mathcal{D}_{A'}(1) \neq \mathbb{B}_k$, which is a contradiction. ■

Remark 3.20. Taking $g = d^2 - 1$, the above proposition is intended to highlight that Theorem 3.17 cannot be used to extend Theorem 2.2 in the natural way to $d^2 - 1$ variables when $d \geq 3$. This is not surprising, as it is known that the state space of M_n for $n > 2$ is not a ball (see [5, §1] for a gentle introduction), in contrast with the Bloch ball of [6] when $n = 2$. Our result thus places more precise dimension bounds on a self-dual set. It remains an open question if there exists $A \in SM_d(\mathbb{C})^g$ such that $\mathcal{D}_A$ is self-dual for some $d \geq 3$ and $g < d^2 - d + 2$.

Thus the question as written is whether there are integers $d \geq 3$ and $g < d^2 - d + 2$, and a tuple $A \in SM_d(\mathbb{C})^g$, for which the free spectrahedron $\mathcal{D}_A$ is self-dual.

Definitions

For a self-adjoint tuple $B = (B_1, \dots, B_g) \in SM_r(\mathbb{C})^g$, its monic pencil at a self-adjoint matrix tuple $X = (X_1, \dots, X_g) \in SM_n(\mathbb{C})^g$ is

$$L_B(X) = I_{rn} - \sum_{j=1}^g B_j \otimes X_j, \quad \mathcal{D}_B(n) = \{X : L_B(X) \succeq 0\}.$$

The free polar dual K° depends only on the matrix convex set K ; hence equal free spectrahedra have equal polar duals.

Let

$$P_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad P_3 = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}.$$

Evert–Passer prove in their Theorem 2.2 that the Pauli free spectrahedron is self-dual:

$$\mathcal{D}_P = \mathcal{D}_P^\circ. \tag{1}$$

Their proof identifies $\mathcal{D}_P$ with the matrix range $\mathcal{W}(P)$ using Choi’s criterion and then invokes the standard free-polar relation $\mathcal{D}_P^\circ = \mathcal{W}(P)$ [1].

Proof intuition

The size d of a coefficient tuple is not intrinsic unless a minimality condition is imposed. Appending a zero reducing summand to every coefficient does not change any positivity inequality: it merely appends a fixed identity block to the monic pencil. We therefore lift the paper’s 2×2 self-dual Pauli example to every coefficient size $d \geq 3$ without changing the free spectrahedron.

Lemma 1 (Zero-padding invariance). *Let $B \in \text{SM}_r(\mathbb{C})^g$ and $d \geq r$. Define $A_j = B_j \oplus 0_{d-r}$. Then $\mathcal{D}_A = \mathcal{D}_B$ at every matrix level. In particular, boundedness and self-duality pass from $\mathcal{D}_B$ to $\mathcal{D}_A$.*

Proof. For every n and every $X \in \text{SM}_n(\mathbb{C})^g$, Kronecker products respect direct sums, so

$$\begin{aligned} L_A(X) &= I_{dn} - \sum_{j=1}^g (B_j \oplus 0_{d-r}) \otimes X_j \\ &= \left(I_{rn} - \sum_{j=1}^g B_j \otimes X_j \right) \oplus I_{(d-r)n} \\ &= L_B(X) \oplus I_{(d-r)n}. \end{aligned}$$

The last matrix is positive semidefinite exactly when $L_B(X)$ is. Hence $\mathcal{D}_A(n) = \mathcal{D}_B(n)$ for every n . Equality of the sets preserves boundedness, and $\mathcal{D}_A^\circ = \mathcal{D}_B^\circ$ by the definition of free polarity. $\square$

Theorem 1 (Affirmative answer to Remark 3.20). *For every integer $d \geq 3$ there exists $A \in \text{SM}_d(\mathbb{C})^3$ such that $\mathcal{D}_A$ is bounded and self-dual. Consequently $g = 3 < d^2 - d + 2$, and the existence question in Remark 3.20 has an affirmative answer as stated.*

Proof. For $j = 1, 2, 3$, set

$$A_j = P_j \oplus 0_{d-2} \in \text{SM}_d(\mathbb{C}).$$

Lemma 1 and (1) give

$$\mathcal{D}_A = \mathcal{D}_P = \mathcal{D}_P^\circ = \mathcal{D}_A^\circ.$$

The set is bounded because $\mathcal{D}_P = \mathcal{W}(P)$ in Theorem 2.2 of [1], and the matrix range of a finite matrix tuple is bounded. Finally, for $d \geq 3$,

$$d^2 - d + 2 \geq 8 > 3 = g.$$

All stated requirements are therefore satisfied (already with $d = 3$). $\square$

Verification and limitations

The proof is an identity at every free level, not a scalar-level argument. A seeded numerical regression check reproduced the block identity for $d = 3, 4, 5$, levels $n = 1, 2, 3$, and ten random self-adjoint triples per pair; the algebraic computation in Lemma 1 is the proof.

The construction has a zero reducing summand. It is therefore not a minimal defining pencil and does not span M_d as an operator system. Remark 3.20 does not impose irreducibility, size-minimality, or a spanning condition, so this is a full answer to its literal statement. The strengthened question under any such nondegeneracy hypothesis remains open here.

Bounded searches through 13 August 2026 used the arXiv id, source title, exact dimension bound, and combinations of “self-dual free spectrahedron”, “Pauli”, “direct sum”, and “zero padding”. No prior statement of this observation was found; arXiv v2 and the May 2026 journal version retain the question. Novelty confidence is moderate pending expert review.

Human-review recommendation. Check only the scope point: whether the authors intended an unstated minimal-pencil convention. Under the definitions and wording actually printed, the block computation is conclusive.

References

- [1] E. Evert and B. Passer, *Matrix convex sets over the Euclidean ball and polar duals of real free spectrahedra*, Linear Algebra Appl. 736 (2026), 54–83; arXiv:2507.20325v2.